



习题6.5

给出在一个整数数组A[1...n]中求出所有元素和的分治算法。

输入：n个元素的数组A[1...n]

输出：数组A[1...n] 总所有元素之和

1. ArraySum(A,1,n)

过程ArraySum(A,low,high)

1. if high – low = 0 return A[low]

2. if high – low > 0

3. mid $\leftarrow \lfloor (low+high)/2 \rfloor$

4. return ArraySum(A,low,mid)+ArraySum(A,mid+1,high)

5. end if

$$T(n) = \begin{cases} \Theta(1) & , n = 1 \\ T(n/2) + T(n/2) & , n > 1 \end{cases}$$

$$T(n) = \Theta(n)$$



```
public static int arraySum(int []A,int low,int high){
    if(high - low == 0)
        return A[low];
    else if(high - low >0){
        int mid = (int)Math.floor((high+low)/2);
        return arraySum(A,low,mid)+arraySum(A,mid+1,high);
    }else
        return Integer.MIN_VALUE;
}
public static void main(String []args){
    int []A={1,2,3,4,5};
    int result = arraySum(A,1,2);
    System.out.println(result);
}
```



习题6.6

整数数组A[1...n], x为一整数。给出一个分治算法, 找出x在A中出现的频度。

输入: 整数数组A[1...n], 整数x

输出: x在数组A[1...n]中出现的频度

1. Frequency(A,1,n,x)

过程Frequency (A,low,high,x)

1. if high - low = 0
2. if A[low] = x return 1
3. else return 0;
4. end if
5. if high - low > 0
6. mid $\leftarrow \lfloor (low+high)/2 \rfloor$
7. return Frequency(A,low,mid,x)+Frequency(A,low+1,high,x)
8. end if

$$T(n) = \begin{cases} \Theta(1) & , n = 1 || n = 2 \\ T(n/2) + T(n/2) & , n > 2 \end{cases}$$

$$T(n) = \Theta(n)$$



```
public static int frequency(int []A,int low, int high,int x){
    if (high - low < 0 )
        return -1;
    else{
        if(high - low == 0 ){
            if (A[low]==x) return 1;
            else return 0;
        }else{
            int mid = (int)Math.floor((high+low)/2);
            return frequency(A,low,mid,x)+frequency(A,mid+1,high,x);
        }
    }
}

public static void main(String []args){
    int []A={7, 5 ,3,2,4,6,7,3,5,7,8};
    int result = frequency(A,0,A.length-1,2);
    System.out.println(result);
}
```



习题6.16

考虑修改MERGESORT算法。将算法应用到输入数组 $A[1...n]$ 上并不断递归调用，直到子问题的规模变得相对很小，即 m 或是小于 m 。此时转向INSERTSORT....。

Algorithm: MERGESORT($A[\text{low}...\text{high}]$)

输入：待排序数组 $A[\text{low},...\text{high}]$

输出： $A[\text{low}...\text{high}]$ 按非降序排列

1. if $\text{low} < \text{high}$ then
2. if $(\text{high} - \text{low} + 1 \leq m)$ then INSERTSORT($A, \text{low}, \text{high}$)
3. else
4. $\text{mid} \leftarrow \lfloor (\text{low} + \text{high}) / 2 \rfloor$
5. MERGESORT($A, \text{low}, \text{mid}$)
6. MERGESORT($A, \text{mid} + 1, \text{high}$)
7. MERGE($A, \text{low}, \text{mid}, \text{high}$)
8. end if
9. end if



分析

$$C(n) = \begin{cases} O(m^2) & , n \leq m \\ 2C(n/2) + n - 1 & , n > m \end{cases} \quad C(m) = m(m-1)/2$$

$$C(n) = 2^k C(n/2^k) + kn - 2^k + 1 \quad \text{见课本p107}$$

$$n/2^k = m$$

$$C(n) = 2^k C(m) + kn - 2^k + 1 = \textcolor{red}{n(m-1)/2} + n \log \frac{n}{m} - \frac{n}{m} + 1$$

$$n(m-1)/2 \leq n \log n \longrightarrow m \leq 2 \log n + 1$$

$$\begin{array}{l} \nearrow m=1 \quad C(n) = n \log n - n + 1 \\ \searrow m=n \quad n(n-1)/2 \end{array}$$



习题6.38

解释当输入已按降序排列时算法QUICKSORT的行为，可以假定输入元素是互不相同的。

Explain the behavior of Algorithm QUICKSORT when the input is already stored in decreasing order. **You may assume** that the input elements are all distinct.

解释当输入已按降序排列时算法**QUICKSORT**的行为(可以假定输入元素是互不相同的)。